

Mark Druffel

Data Scientist & Analytics Engineer



Summary

Versatile data scientist who finds the strategic opportunities hiding in technical work and turns them into a story the business can act on. 10+ years experience in data strategy, analysis, engineering, and project management with a proven track record building scalable platforms, leading teams, and delivering measurable outcomes.

Experience

Lead Data Scientist

84.51° | Remote
Nov '22 - Present

Libraries

kedro, ibis-project,
pyspark, delta,
statsmodels,
scikit-learn, uv,
pandera, pointblank

Other Tools

databricks, medallion
architecture,
unitycatalog, duckdb,
azure, quarto

Lead a data science team building tools and insights behind Kroger Marketing's campaign strategy and execution. Built the case for and delivered a composable measurement pipeline cutting compute costs by \$200K annually and speeding up feature development by 2.5x.

Data Pipeline Modernization Project - designed, pitched, and delivered a fully composable campaign measurement pipeline to deliver uplift and other KPIs at the pace of the business.

- Created and presented the proposal that secured stakeholder alignment and approval.
- Championed the novel use of kedro and ibis to enable composable, local development improving the team's new-feature velocity by 2.5x.
- Implemented dynamic predicate pushdown, reducing runtimes by 9x and Databricks compute costs by \$200K annually.
- Designed a layered architecture enabling plug-and-play model prototyping, enabling the team to develop new uplift models that drove 4x higher household penetration and uplift.

Agentic Infrastructure Development - developed a suite of Claude skills and agents to assist with pipeline development and operations lowering the learning curve and improving quality.

Campaign Management App Design - partnered with engineering to create data model design.

- Led stakeholder interviews to understand requirements.
- Designed recursive campaign hierarchy, unlocking several feature requests.

Senior Data Analyst

DocStation | Remote
Apr '21 - Nov '22

Libraries

shiny, tidyverse,
pyspark, leaflet, sf

Other Tools

airflow, superset, aws
snowflake, docker

Founding member of a startup enabling pharmacists to deliver value-based care. Led all analytics initiatives and contributed to data engineering infrastructure. Built pharmacy targeting dashboard that increased signups by 15%.

New User Targeting Dashboard - designed and developed a full-stack dashboard enabling product to identify and prioritize new pharmacy targets, increasing pharmacy signups by 15%.

- Built a nationwide pharmacy dataset from open-source data, merged it with internal records, and augmented it with demographic and geospatial features to enable targeting metrics.
- Developed an interactive dashboard used for exploring a map of target pharmacies, a sortable metrics table, and dynamic cross filtering.

Patient Targeting Analysis - performed recurring targeting analyses that improved the platform's targeting model and surfaced ad hoc targeting opportunities.

Analytics Consultant

Propeller |
Portland, OR
Jan '17 - Apr '21

Led data strategy, analysis, architecture, BI, and development projects for clients including Nike, Salesforce, KinderCare, Cambia, and Mercy Corps. Built an internal analytics practice that saved \$250K annually and delivered reporting and analytics solutions to the company.

Internal Analytics Practice - created an internal analytics practice to refine implementation plans for client work by building a suite of tools including a consultant-utilization Slackbot and a data-driven recruiting process, improving recruiting quality and saving \$250K annually.

Libraries tidyverse, dbplyr, shiny, tidytext, prophet, fable, d3.js Other Tools rmarkdown, Powerpoint, aws, databricks, Neo4j, sqlite, slackbot, Invision	User Feedback Pipeline - transformed a weeks-long, static reporting process into a real-time feedback loop for a mobile app team, enhancing their ability to incorporate user behavior and feedback into their priorities. Built API wrappers to collect and merge usage, issue, and survey data, and an NLP model to mine survey responses to generate new attributes for metrics. Delivered a dashboard with KPIs and exploratory analytics tools. Data-Driven Recruiting Prototype - designed a prototype recruiting dashboard to realize an SVP's data-driven recruiting vision. Led org-wide stakeholder interviews to map process and deliver change management. Designed and developed a data model that merged several systems to provide an end-to-end picture of the recruiting pipeline and KPIs. Designed and developed a functional prototype dashboard with drill-down by discipline, team and recruiter, improving transparency and communication throughout the org. Analytics Workshops - developed and facilitated PMO workshops on how analytics projects differ from software projects, improving their analytics project planning capability.
Reporting Manager Northern Trust Chicago, IL Oct '15 - Dec '16 Libraries pandas, d3.js, stats, ggplot2, xml2 Team Lead Sept '13 - Oct '15 Libraries pandas Tools Excel, Tableau	Managed a new product team, formed in response to my prior systems analysis, building tools, infrastructure, and strategy for the business intelligence operations team. Regression Testing Tool - designed and developed a Python tool to automate data model regression testing, reducing release issues by 4x within six months. Nextgen Reporting Application - designed and developed an MVP SaaS business intelligence application with pre-built reports for hedge fund managers using d3.js and React. Legacy System Monitoring Tools - designed and developed telemetry-based monitoring system with KPIs and demand forecasting, enabling hardware planning, tech debt management, and new client onboarding without unexpected slowdowns. Led a global team of business intelligence and operations analysts responsible for data modeling, reporting, and access management across the entire platform, serving internal users and clients with over \$200B in AUM. Reporting Systems Analysis Project - spearheaded a holistic systems analysis of log data, systems metadata, and process data that identified architectural issues, pipeline bottlenecks, and process inefficiencies. Documented findings in a report that led management to create a new analytics product team.
Audit Jun '10 - Sept '13 Tools Excel, vba	Senior Auditor - Plante Moran Chicago, IL Mar '13 - Sept '13 Staff Auditor - Plante Moran Chicago, IL Jun '11 - Mar '13 Staff Auditor - Fifth Third Bank Cincinnati, OH Jun '10 - Jun '11 Started career in audit. Developed understanding of financial reporting, internal controls, process analysis, and stakeholder communication. Education
MSBA Notre Dame BA Ohio University Hobbies	Notre Dame (2016) Master's in Business Analytics, coursework in data management, statistics, operations research, and machine learning Ohio University (2010) Bachelor of Business Administration, Triple Major Accounting, Finance, & Management Info Systems Open-Source Work
Rock climb, bike, DIY, read, travel, games	Contributions to the kedro, ibis-project, and pointblank repos, and authored the harvestR and bambooR repos.